



Source-Free Domain Adaptation with Vision-Language Prior

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Abstract

Source-Free Domain Adaptation (SFDA) seeks to adapt a source model, which is pre-trained on a supervised source domain, for a target domain, with only access to unlabeled target training data. Relying on pseudo labeling and/or auxiliary supervision, conventional methods are inevitably error-prone. To mitigate this limitation, in this work we for the first time explore the potentials of off-the-shelf vision-language (ViL) multimodal models (e.g., CLIP) with rich whilst heterogeneous knowledge. We find that directly applying the ViL model to the target domain in a zero-shot fashion is unsatisfactory, as it is not specialized for this particular task but largely generic. To make it task-specific, we propose a novel DIFO++ approach. Specifically, DIFO++ alternates between two steps during adaptation: (i) Customizing the ViL model by maximizing the mutual information with the target model in a prompt learning manner, (ii) Distilling the knowledge of this customized ViL model to the target model, centering on gap region reduction. During progressive knowledge adaptation, we first identify and focus on the gap region, where enclosed features are entangled and class-ambiguous, as it often captures richer task-specific semantics. Reliable pseudo-labels are then generated by fusing predictions from the target and ViL models, supported by a memory mechanism. Finally, gap region reduction is guided by category attention and predictive consistency for semantic alignment, complemented by referenced entropy minimization to suppress uncertainty. Extensive experiments show that DIFO++ significantly outperforms the state-of-the-art alternatives. Our code and data are available at <https://github.com/tntek/DIFO-Plus>.

Keywords Domain adaptation · Source-free · Vision-language prior · Multimodal foundation model customization · Gap region reduction

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1 Introduction

Unsupervised Domain Adaptation (UDA) relies on both well-annotated source data and unannotated target data. Such a paradigm is particularly crucial in applications like autonomous driving (e.g., domain-adaptive semantic segmentation (Chen et al., 2023, 2024c)) and medical image analysis (e.g., cross-modality diagnosis (Wang et al., 2024)). However, due to heightened safety and privacy concerns, accessing source data freely has become difficult (Liang et

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al., 2019; Lao et al., 2021). In response, Source-Free Domain Adaptation (SFDA) has gained attention as a more practical solution, aiming to transfer a pre-trained source model to the target domain using only unlabeled target data.

Due to the absence of source samples, traditional distribution matching approaches are no longer viable (Ganin & Lempitsky, 2015; Kang et al., 2019). The predominant alternative is self-supervised learning, which generates or mines auxiliary information to facilitate unsupervised adaptation. Two main approaches exist: constructing a pseudo source domain to leverage established UDA methods such as adversarial learning (Xia et al., 2021; Kurmi et al., 2021) or domain shift minimization based on distribution measurement (Ding et al., 2022; Tian et al., 2021; Kundu et al., 2022; Zhou et al., 2024) and mining extra supervision from the source model (Lao et al., 2021; Wang et al., 2022; Huang et al., 2021) or target data (Yang et al., 2022; Tang et al., 2024; Wang et al., 2025). In the presence of domain distribution shift, applying the source model to the target domain introduces inevitable errors in pseudo-labeling or auxiliary supervision, thereby limiting adaptation performance.

To address identified limitations, we pioneer the exploration of off-the-shelf multimodal foundation models, such as the vision-language (ViL) model CLIP (Radford et al., 2021), transcending the constraints of both the source model and target data knowledge. However, direct application of the ViL model proves unsatisfactory, lacking specialization for specific tasks. To tackle this problem, in our preliminary conference work (Tang et al., 2024), we propose a novel distillation approach named *Distilling multiModal Foundation model* (**DIFO**). Figure 1 demonstrates the idea of DIFO. Initially, DIFO customizes the ViL model through unsupervised prompt learning to incorporate task-specific information. Since both the ViL model and the source model struggle

to make accurate predictions in a target domain, we adopt unbiased mutual information maximization to regulate this prompt learning. Consequently, DIFO distills the knowledge from this customized ViL model to the target model, with joint supervision through two designed regularization terms. The first strategy, most-likely category encouragement, prompts the model to concentrate on the most likely classes, thus facilitating coarse-grained distillation. In contrast, the second strategy, predictive consistency, ensures fine-grained distillation by maintaining consistency in predictions.

We observe that training samples contribute differently to expressing task specificity during model adaptation. In particular, easy samples offer limited new information, whereas task-specific knowledge is often concentrated in hard samples, many of which reside in the *gap region* characterized by uncertainty and class ambiguity. However, the DIFO approach treats all samples equally during training. This uniform attention strategy may cause DIFO to overlook valuable information hidden within the gap region, especially as the proportion of easy data increases.

Therefore, we propose DIFO++, an enhanced SFDA framework with a vision-language prior, building upon DIFO. Specifically, in the phase of distilling customized knowledge (Fig. 1), we propose a novel gap region-driven scheme, allowing the distillation to occur through a process of reducing the gap region. To address this, we first identify the gap region using a novel metric named referenced entropy, which captures uncertainty and class ambiguity in feature space. We then assign pseudo-labels to the uncertain region by fusing historical cross-model predictions. To progressively reduce the gap region, we introduce three regularization strategies: *Category attention calibration*, *Predictive Consistency*, and *Gap region compression*, which collectively drive the features in the gap region toward more discriminative and class-consistent areas.

Beyond DIFO (Tang et al., 2024) published in CVPR24, we introduce several key differences with DIFO++: (1) Additionally considering data uncertainty to enhance distilling of a frozen ViL model. (2) Extending the DIFO framework by introducing a new gap region-driven progressive distillation strategy. This approach ensures that adaptation focuses on semantically rich, hard-to-classify regions rather than treating all samples uniformly. (3) Providing more experimental analysis and discussion, including the rationale analysis of evolving DIFO to the DIFO++ model.

To sum up, we make these **contributions** below:

- (1) Pioneering the use of generic but heterogeneous knowledge sources (e.g., the off-the-shelf ViL model) for the SFDA problem, transcending the limited knowledge boundary of a pre-trained source model and unlabeled target data.

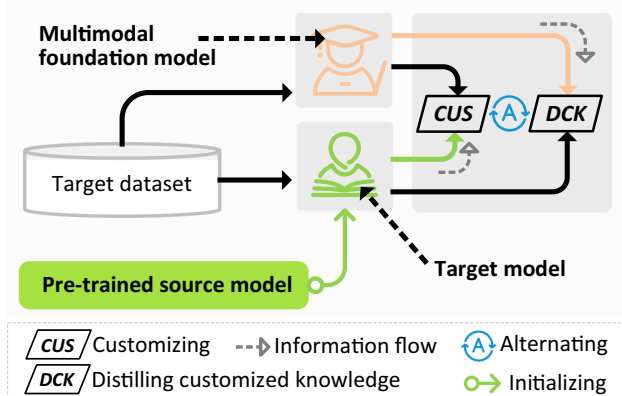


Fig. 1 We expand beyond traditional SFDA methods that rely solely on a pretrained source model and unlabeled target data. Instead, we innovate by exploring off-the-shelf multimodal foundation models, such as CLIP, in an unsupervised manner (marked by the box with blue background)

- (2) Development of a novel DIFO++ approach to effectively distill useful task-specific knowledge from the general-purpose ViL model. Specifically, our method leverages referenced uncertainty to identify and rank gap region samples by their difficulty, enabling the model to progressively distill knowledge from increasingly informative and task-specific examples.
- (3) Extensive evaluation on standard benchmarks, demonstrating the significant superiority of our DIFO++ over previous state-of-the-art alternatives under conventional closed-set settings, as well as more challenging partial-set and open-set settings.

2 Related Work

2.1 Source-Free Domain Adaptation

Existing SFDA approaches fall into three distinct categories. The first explicitly aligns the pseudo source domain with the target domain, treating SFDA as a specialized case of unsupervised domain adaptation. This alignment is achieved by constructing the pseudo source domain through a generative model (Tian et al., 2021) or by splitting the target domain based on prior source hypotheses (Du et al., 2021).

The second group extracts cross-domain factors from the source domain and transfers them via successive model adaptation to align feature distributions across the two domains. For example, (Tang et al., 2019) establishes a mapping relationship from a sample and its exemplar Support Vector Machine (SVM) (an individual classifier) on the source domain to ensure individual classification on the target domain. Some approaches leverage pre-trained source models to generate auxiliary factors, such as multi-hypothesis (Lao et al., 2021), prototypes (Tanwisuth et al., 2021; Zhou et al., 2024), source/target distribution estimation (Ding et al., 2022; Lee et al., 2022), or hard samples (Li et al., 2021) to aid in feature alignment.

The third group incorporates auxiliary information refined from the unlabeled target domain. In addition to widely used pseudo-labels (Liang et al., 2020; Chen et al., 2022; Rai et al., 2025), prediction confidence (Wang et al., 2025; Xu et al., 2025), geometry information, such as intrinsic neighborhood structure (Litrico et al., 2023), hyper-graph connection (Hwang et al. 2024; Jiang et al. 2025), and target data manifold (Tang et al. 2022, 2024), has also been exploited.

Despite continual advancements, these methods are limited by the knowledge derived solely from the pretrained source model and unlabeled target data. We break this limitation by tapping into the rich knowledge encoded in off-the-shelf multimodal foundation models.

2.2 Uncertainty-Aware Domain Adaptation

Within the Domain Adaptation (DA) context, uncertainty serves as a critical extrinsic manifestation of domain shift. Consequently, uncertainty-aware approaches have long been a hallmark of DA research. While these methods share a common philosophy of easy-to-hard adaptation, their specific methodological distinctions emerge in the following two aspects:

(1) *In terms of the mechanism for identifying hard samples*, existing work is in two lines. The first one is output-based methods using standard entropy (Bishop, 2006), loss values (Yang et al., 2023; Yu et al., 2025), energy-based scores (Liu et al. 2020), and class margin (Song et al. 2024). The second one is the characteristic-based methods that leverage geometric feature structures (Zhu et al. 2022; Yang et al. 2021) or historical information (Lai et al., 2024). For instance, (Zhu et al., 2022) utilizes sliding window voting to propagate labels and distinguish between easy and hard samples. Alternatively, (Yang et al., 2021) employs local affinity for sample mining, and (Lai et al., 2024) characterizes hard samples via sub-prototypes derived from historical feature memory.

(2) *In terms of the strategy for exploiting the mined hard data to facilitate adaptation*, existing strategies generally fall into three categories. The first follows a divide-and-conquer paradigm: (Zhu et al., 2022) applies pseudo-labels to easy samples while using adversarial learning to enforce hard-to-easy feature alignment; (Yu et al., 2025) employs KL-regularized weighting to align target data with the source domain; and (Song et al., 2024) focuses exclusively on high-confidence samples, effectively discarding hard data. The second category (Yang et al., 2023; Shin et al., 2020) reformulates the adaptation task as semi-supervised learning, treating easy-sample pseudo-labels as ground truth. The third category leverages consistency learning to drive feature clustering-based adaptation: (Lai et al., 2024) utilizes sub-prototypes and cycle-consistent matching to classify private target classes, while (Yang et al., 2021) enforces label consistency among hard samples with high local affinity.

In summary, regarding the two methodological dimensions above, DIFO++ diverges significantly from existing approaches:

- (1) The proposed referenced entropy explicitly leverages the initial semantic cues provided by the source model for the target domain. This addresses a critical gap in existing approaches, which often ignore this inherent prior information in favor of generic uncertainty metrics.
- (2) Our curriculum-driven distillation employs an unbiased alignment anchored by mutual information entropy regularization. By avoiding absolute reliance on specific subsets, this design circumvents confirmation bias and error propagation. In contrast, existing methods' over-

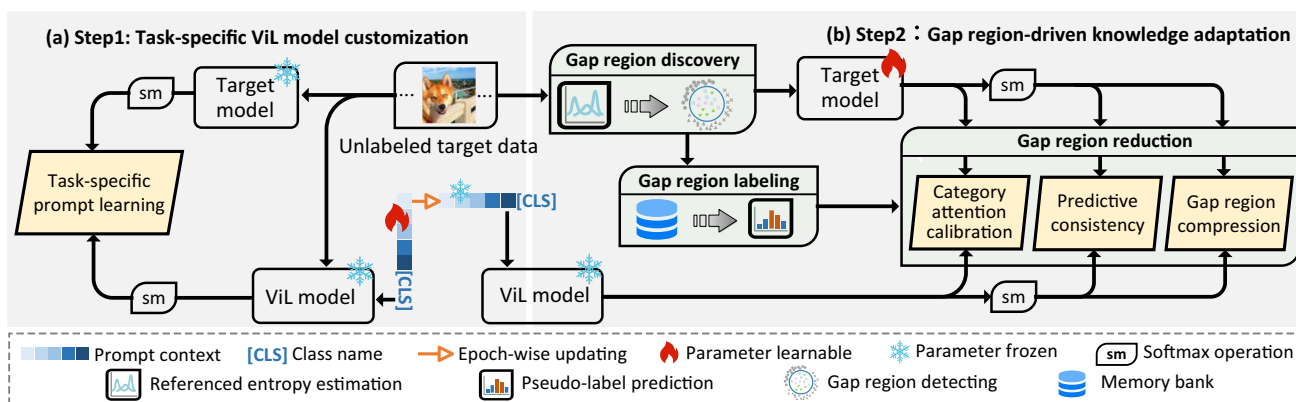


Fig. 2 Overview of DIFO++: Our method alternates between two steps. First, we perform **a** *task-specific customization* (Sect. 3.1) of a ViL model through task-specific prompt learning ($\mathcal{L}_{S-1}$). This is achieved under soft predictive guidance using mutual information maximization. Second, we perform **b** *gap region-driven knowledge adaptation* (Sect. 3.2), which recasts knowledge transfer as the task of shrinking the gap region, in which feature samples are entangled and exhibit high class ambiguity. To this end, we first identify the gap region using

referenced entropy, and then label the gap region by fusing the predictions of the target and ViL models that are stored in a memory bank. Finally, we reduce the gap region by incorporating regularizations: Category attention calibration ($\mathcal{L}_{cac}$), Predictive consistency ($\mathcal{L}_{pc}$), and Gap region compression ($\mathcal{L}_{rc}$). This pipeline is structured into three functional modules: (1) Gap region discovery (Sect. 3.2.1), (2) Gap region labeling (Sect. 3.2.2), and (3) Gap region reduction (Sect. 3.2.3)

dependence on easy samples results in a restrictive unidirectional alignment, where labeling errors in easy data are easily amplified.

2.3 Large Multimodal Models

Multimodal vision-language (ViL) models, such as (CLIP Radford et al. 2021) and (OpenCLIP Cherti et al. 2023), have shown promise across various mono-modal and multimodal tasks by capturing modality-invariant features. Approaches in this domain can be broadly categorized into two lines. The first line focuses on enhancing ViL model performance. For instance, in (Zhou et al., 2022); (Ge et al., 2025), prompt learning optimizes the text encoder through the use of tailored, learnable prompts designed for specific scenarios. Other efforts aim to improve data efficiency by repurposing noisy data (Andonian et al., 2022) or by leveraging multimodal synthetic data (Li & Li, 2025).

The second line utilizes ViL models as external knowledge to enhance downstream tasks, as demonstrated in this paper. Previous work in knowledge transfer primarily falls into two frameworks. For the first scheme, where the ViL model is directly applied to the target task in a zero-shot fashion (Liang et al., 2023), domain generality is leveraged without task-specific refinement. The second scheme does not focus on source model adaptation. Instead, it fine-tunes the ViL model (Li et al., 2025) to the target domain through prompt or adaptor learning with an amount of manual labels (Cho et al., 2023).

Most recently, multimodal large models such as CLIP and ShareGPT (Chen et al., 2024b) have been introduced

to advance SFDA, substantially enhancing adaptation performance by leveraging generic knowledge in these models. **Three orthogonal strategies** are adopted: (1) leveraging knowledge encoded in a single foundation model via knowledge adaptation (Tang et al., 2024; Zhang et al., 2025), (2) integrating multi-LLM-teacher guidance via curriculum learning (Chen et al., 2024a), and (3) denoising ViL predictions (Tang et al., 2025). Our DIFO++ aligns with the first strategy.

Among previous work, the most relevant work to DIFO++ is Co-learn-V (Zhang et al., 2025). Specifically, in each adapting epoch, DIFO++ first tailors CLIP’s generic knowledge to the specific task, and then performs gap-region-driven knowledge distillation that explicitly accounts for data uncertainty during adaptation. Co-learn-V adopts a prototype-guided approach where zero-shot CLIP predictions are injected to refine the target prototypes. Two key features distinguish DIFO++ from Co-learn-V: (1) a new customize–then–distill paradigm rather than a conventional prototype-based scheme integrating the original CLIP knowledge, and (2) explicit modeling of data uncertainty, which Co-learn-V does not consider.

3 Methodology

Problem statement We have two distinct yet related domains, the source and target domains, characterized by the same C categories. The source domain is represented by a source model $\theta_s : \mathcal{X}_s \rightarrow \mathcal{Y}_s$ pretrained on a labeled training data $\mathcal{D}_s = \{\mathcal{X}_s, \mathcal{Y}_s\}$. For the target domain, we only have n unlabeled

beled training data $\mathcal{X}_t = \{\mathbf{x}_i\}_{i=1}^n$ without any category labels. The objective is to derive a target model $\theta_t : \mathcal{X}_t \rightarrow \mathcal{Y}_t$ by adapting the source model θ_s on $\mathcal{X}_t$. To that end, we explore the potential of a Visual-Language (ViL) model θ_v .

Overview of DIFO++ To address SFDA with a ViL model, we propose the DIFO++ framework, as depicted in Fig. 2. Our method alternates between two distinct stages to customize and adapt rich ViL knowledge reliably during adaptation. In the *first* stage, we customize the ViL model via prompt learning. This serves to suppress the guidance errors from the ViL model.

In the *second* stage, we perform gap region-driven knowledge adaptation, which frames the knowledge transfer as a process of reducing the gap region where the enclosed features are highly entangled and class-ambiguous. This process begins by identifying gap regions, whose difficulty gradually increases, using the proposed metric of referenced entropy. Next, the identified gap regions are labeled through a fusion of predictions from the target model and the customized ViL models, assisted by a memory bank that retains historical information. Subsequently, we reduce the gap region by adapting knowledge between the target model and the customized ViL models. This adaptation is guided by a tailored constraint that promotes the selection of the most probable category labels in the logit space, while simultaneously preserving predictive consistency. Additionally, gap region compression regularization further strengthens the adaptation by minimizing referenced uncertainty.

3.1 Stage I: ViL Model Customization

We adopt the prompt learning framework for ViL model customization, with ViL's parameters all frozen throughout. The only learnable part in customization is the prompts each assigned for a specific class. To optimize these prompts, we need a quality supervision. In SFDA, however, it is challenging to customize such a domain-generic ViL model towards the target domain, at the absence of a well-trained target domain model. This is because, none of them can reasonably make predictions. That means there is no clearly preferred supervision signals available.

To address this challenge, we propose to explore the wisdom of the crowd by leveraging their predictive interaction as more reliable supervision. Formally, given an unlabeled target sample $\mathbf{x}_k$, we denote the predictions by the target model and the ViL model as $\theta_t(\mathbf{x}_i)$ and $\theta_v(\mathbf{x}_i)$, respectively. We conduct the ViL model customization by maximizing the mutual information of the two predictions as:

$$\mathcal{L}_{S-I} = -\min_v \mathbb{E}_{\mathbf{x}_i \in \mathcal{X}_t} I(\theta_t(\mathbf{x}_i), \theta_v(\mathbf{x}_i, \mathbf{v})) \quad (1)$$

where $\mathbf{v}$ is the prompt context to be learned and the function $I(\cdot, \cdot)$ measures the mutual information (Ji et al., 2019).

This alignment design differs significantly from the conventional adoption of the Kullback–Leibler (KL) divergence. First, mutual information is a lower optimization bound than KL divergence, facilitating deeper alignment (see Lemma 1 which can be recalled from (Cover & Thomas, 2006)).

Lemma 1 *Given two random variables X, Y . Their mutual information $I(X, Y)$ and KL divergence $D_{KL}(X||Y)$ satisfy the unequal relationship as follows.*

$$-I(X, Y) \leq D_{KL}(X, Y). \quad (2)$$

Remark We presented this lemma to provide a self-contained theoretical motivation for using mutual information in our context, rather than claiming it as an original mathematical contribution.

Crucially, the KL divergence exhibits an inherent bias towards a specific prediction, making it less suitable for our context where none of the predictions holds a clear advantage. On the contrary, mutual information considers the joint distribution or correlation between the two predictions. This distinction arises from their respective definitions: $-I(X, Y) = -H(X) + H(X|Y)$ and $D_{KL}(X, Y) = -H(X) + H(X : Y)$, where

$$\begin{aligned} H(X|Y) &= -\sum p(\mathbf{x}, \mathbf{y}) \log p(\mathbf{x}|\mathbf{y}) \\ H(X : Y) &= -\sum p(\mathbf{x}) \log p(\mathbf{y}). \end{aligned} \quad (3)$$

3.2 Stage II: Gap Region-Driven Knowledge Adaptation

At this stage, we adapt the customized ViL knowledge by explicitly reducing the gap region. This strategy is motivated by an empirical observation of the target feature distribution under the source model's mapping. As illustrated in Fig. 3 (a), the feature space presents a dual structural pattern: (1) class-consistent, island-like clusters characterized by high intra-class compactness, and (2) transitional regions situated between these clusters, where features are entangled and class-ambiguous. These transitional areas often give rise to model uncertainty and misclassification due to their semantic ambiguity. Given their role in separating well-formed class clusters, we refer to these regions as *gap regions*.

In practice, we first identify the region and subsequently assign pseudo-labels to it. With the resulting pseudo semantics, the gap region is progressively reduced. The following subsections provide an elaboration on this process.

3.2.1 Gap Region Discovery

The most salient characteristic of the gap regions is the intermixing of data points from multiple classes, which

is typically reflected in high uncertainty. For the issue of gap region discovery, an uncertainty metric capable of achieving high discriminability plays a fundamental role. Existing approaches often use the absolute value of the signal/prediction's attribution, such as entropy (Bi et al., 2022), energy (Banitalebi-Dehkordi et al., 2023), and cluster density (Choi et al., 2019), to evaluate data uncertainty. They implicitly reference a uniform probability distribution as the baseline. However, in the SFDA context, the learning process begins with a non-uniform probability distribution, as defined by the pre-trained source model. Thus, the previous metrics are not suitable for SFDA.

Remark Although prior theories suggest that entropy-based uncertainty can be brittle (Gal & Ghahramani, 2016; Gawlikowski et al., 2023; Maddox et al., 2019), these critiques are grounded in supervised learning. This diverges from the SFDA setting (lacking labels and involving domain shift), rendering such classical insights less applicable. We therefore follow the established paradigm of using information entropy as a viable proxy for uncertainty in SFDA (Lee et al., 2024; Safaei et al., 2025; Xu et al., 2024).

To address this issue, we propose a novel metric, *referenced entropy*. For a specific training sample, its semantic attributes, such as category, foreground, background, and classification difficulty, are fixed. A learned well model should map these fixed attributes to stable outputs. In this view, we regard high uncertainty as significant differences in prediction, correspondingly reflected as vibration in terms of information entropy. Inspired by this, we introduce the referenced entropy, specifically designed to measure entropy variation relative to an evolving, adaptation-aware probability distribution reference. This design incorporates both the source model and the adaptation process.

Assume the gap region is collectively denoted by $\mathcal{R}_t \subset \mathcal{X}_t$. With the proposed referenced uncertainty, we accomplish the gap region discovery, as outlined below.

(a) *Referenced entropy estimation* First, we derive an evolving adaptation-aware reference, formulated as:

$$\rho_i^k = \rho_i^{k-1} + \delta h_i^k, \quad k \in [0, K-1], \quad (4)$$

where δ is a weighting parameter; K is the number of epochs, and h_i^k is the entropy of the prediction by the current in-training model.

Each sample's reference ρ_i^k is initiated (when $k = 0$) by calculating the entropy of the source model's prediction. As in Eq. (4), the reference updates in the Exponential Moving Average (EMA) fashion, providing accumulative uncertainty estimation by continuously incorporating the adaptation dynamics over epochs.

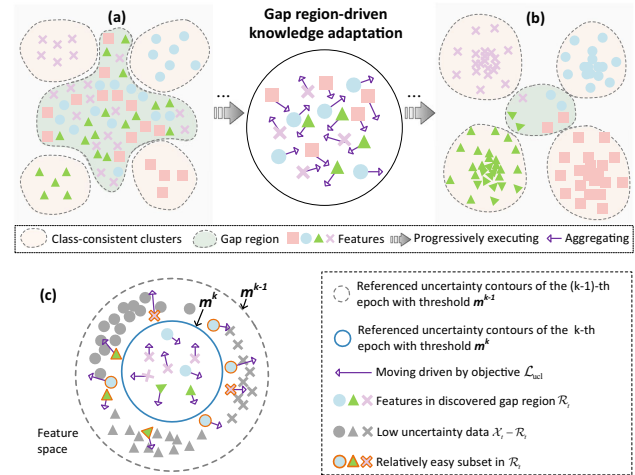


Fig. 3 An illustrative idea of gap region reduction. **a** The gap region refers to the transitional area between class-consistent clusters, detected by the proposed metric of referenced entropy. **b** Through a gap region-driven knowledge adaptation, features within the gap region are properly consolidated, reducing its extent. **c** A snapshot of this progressive adaptation, in which $\mathcal{L}_{\text{re}}$ guides the movement of features within the gap region $\mathcal{R}_t$, ultimately resulting in gap region compression

With ρ_i^k , the referenced uncertainty of any target sample $x_i \in \mathcal{X}_t$ in the k -th epoch is then measured as:

$$U(x_i) = H(\theta_t(x_i)) - \rho_i^k, \quad (5)$$

where $H(\theta_t(x_i))$ returns the predictive entropy of x_i with the current target model θ_t .

(b) *Detecting gap region* To achieve a gradual discovery process, we introduce an uncertainty threshold $m^k, k \in [0, K-1]$ that increases epoch-wise to schedule the adaptation process. For each training iteration in the k -th epoch, we split the hard set from the entire target data, denoted by $\mathcal{R}_t$, which consists of the data whose uncertainty is larger than the current epoch's threshold m^k . Formally,

$$\mathcal{R}_t = \left\{ x_i \mid U(x_i) > m^k \right\}_{i=1}^{|\mathcal{R}_t|}, \quad m^k = \epsilon \cdot \gamma^k, \quad (6)$$

where $\epsilon \in (0, 1)$ and $\gamma > 1$.

The discovery rule in Eq. (6) suggests that by increasing m^k for $0 \leq k \leq K-1$, training can begin with a subset that includes all potential hard samples. As training progresses, samples are gradually excluded based on their associated uncertainty, following a trajectory from low to high uncertainty. This strategy leads to a progressive increase in the difficulty level of the discovered gap regions.

3.2.2 Gap Region Labeling

In practice, we achieve the gap region labeling by assigning confused pseudo-labels to the samples contained in it. This

process builds upon a memory bank, as shown in Fig. 2 (b). This memory bank manages two types of knowledge for all target samples: (1) predictions by the target model: $\{\mathbf{p}'_i\}_{i=1}^n$, and (2) predictions by the ViL model: $\{\mathbf{p}''_i\}_{i=1}^n$.

Throughout the adaptation, the predictions from the target model are updated iteratively. At the end of each training iteration, the newly predicted labels for the training batch from the target model replace their counterparts in the prediction bank. In contrast, predictions from the ViL model are updated collectively in an epoch-wise manner, triggering updates every M iterations. This mixed-update strategy is designed to strike a balance between maintaining the stability of the customized ViL model's guidance and capturing the task-specific dynamics inherent in the adaptation.

Based on the stored prediction information, the creation of a pseudo-label for any samples in the gap region $\mathcal{R}_t$ involves a historical prediction fusion process as:

$$\bar{\mathbf{p}}_i = \tau \mathbf{p}'_i + (1 - \tau) \mathbf{p}''_i, \quad (7)$$

where the weight τ , drawn from an Exponential distribution with parameter λ , is a crucial factor.

The rationale behind incorporating this confusion is to harness the collective knowledge of both the target model and the ViL model in order to enhance the discernment of probable category labels for each sample. Additionally, this fusion introduces dynamic bias rectification (represented by $\mathbf{p}_i$) based on the guidance from the customized ViL model ($\mathbf{p}''_i$). The role of $\mathbf{p}'_i$ is to provide adjustments, leading us to adopt an asymmetric random weighting approach represented by τ .

3.2.3 Gap Region Reduction

From the perspective of feature distribution, reducing the gap region relies on guiding the samples within it toward class-consistent, island-like clusters. To this end, we introduce three key designs. Specifically, *category attention calibration* and *predictive consistency* jointly encouraging this sample migration by adapting quality semantic knowledge. Moreover, *gap region compression*, achieved through uncertainty minimization, further reinforces this process. The details of these designs are presented below.

(a) *Category attention calibration* The gap region labeling fuses the semantics from the target and ViL model (see Eq. (7)). Since the prediction errors of both models are not strictly constrained, the fused pseudo-label does not guarantee that the highest-probability category is reliable. Nevertheless, the incorporation of multi-perspective knowledge significantly improves the likelihood that the top-N predicted categories include the ground truth.

Inspired by the above insight, we formulate a regularization to guide the target model's focus toward these

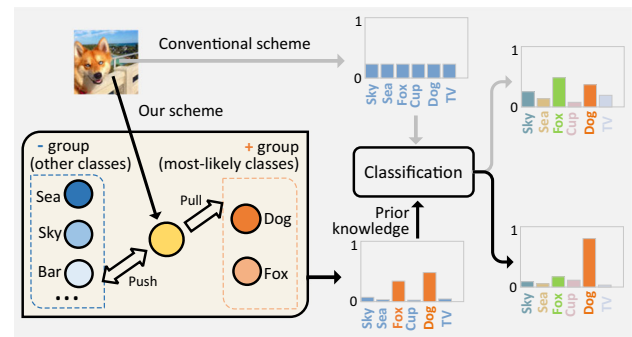


Fig. 4 Illustration of the design of category attention calibration. In contrast to the conventional approach that assigns equal importance to all categories (depicted by the gray line), our approach (represented by the black line) introduces additional supervision by incorporating extra knowledge about the two most likely categories

categories, referred to as *category attention calibration*. Figure 4 illustrates this regularization. Specifically, we begin by identifying the *top-N* most probable categories using $\bar{\mathbf{p}}_i$. The indices of these identified categories are denoted by $\mathcal{M}_i = \{m_k\}_{k=1}^N$. With $\mathcal{M}_i$, the target model's logit of a target domain sample x_i , denoted as $\mathbf{l}_i$, is segregated into positive and negative category groups. We can define this regularization as:

$$\mathcal{L}_{\text{cac}} = \min_{\theta_t} \mathbb{E}_{x_i \in \mathcal{R}_t} \log \frac{\exp(a_i/\iota)}{\sum_{j \neq \mathcal{M}_i} \exp(b_i \cdot \mathbf{l}_{i,j}/\iota)} \quad (8)$$

$$a_i = \prod_{k=1}^N \mathbf{l}_{i,m_k}, \quad b_i = \sum_{k=1}^N \mathbf{l}_{i,m_k}$$

where $\mathbf{l}_{i,a}$ denotes the a -th element of $\mathbf{l}_i$, ι is the temperature parameter, and $\mathcal{R}_t$ is the gap region identified by Eq. (6).

In Eq. (8), we note that the product operation with a_i in the numerator amplifies penalties for the probability decrease on the most likely categories compared to the sum form. Similarly, the sum with b_i in the denominator serves as an increasing weighting parameter to enhance suppression of values at other locations. Moreover, a_i is more sensitive to changes than b_i due to $\frac{\partial a_i}{\partial m_k} \propto O(n^{N-1})$ and $\frac{\partial b_i}{\partial m_k} \propto O(1)$. By combining the use of a_i and b_i , we globally impose a calibration effect on the elements corresponding to the most likely categories within the logit $\mathbf{l}_i$. Essentially, attention is introduced to these potential categories, as illustrated in the box with a yellow background in Fig. 4.

(b) *Predictive consistency* For the purpose of knowledge adaptation, we incorporate the conventional predictive consistency loss as:

$$\mathcal{L}_{\text{pc}} = \min_{\theta_t} [-\mathbb{E}_{x_i \in \mathcal{R}_t} \mathbf{I}(\theta_t(\mathbf{x}_i), \theta_v(\mathbf{x}_i, \mathbf{v}*)) + \alpha \mathcal{L}_b], \quad (9)$$

where $\mathcal{R}_t$ is the gap region, $\theta_t(\mathbf{x}_i)$ represents the target prediction, $\theta_v(\mathbf{x}_i, \mathbf{v})$ denotes the ViL prediction, and $\mathbf{v}$ is

Algorithm 1: Training procedure of DIFO++

Input: Pre-trained source model θ_s , target model θ_t , ViL model θ_v , unlabelled target domain $\mathcal{X}_t$, learnable prompt context v , epoch number K , iteration number per epoch M .

Output: The adapted target model θ_t .

- 1 **Initialisation:** Set model $\theta_t = \theta_s$, prompt $v = "a photo of a [CLS]."$
- 2 **for** $k = 1$ **to** K **do**
- 3 Update ViL predictions in the prediction bank;
 /* -- Step1: customization -- */
- 4 **for** $j = 1$ **to** M **do**
- 5 Sample a batch from $\mathcal{X}_t$;
- 6 Forward prompt v , this batch $\mathcal{X}_t^b$ by θ_v ;
- 7 Forward $\mathcal{X}_t^b$ data through θ_t ;
- 8 Customize θ_v by optimizing $\mathcal{L}_{S-I}$ (Eq. (1)) and obtain task-specific prompt context v^* .
- 9 /* -- Step2: adaptation -- */
- 10 Compute uncertainty margin m^k .
- 11 Compute adaptation-aware reference ρ^k .
- 12 **for** $j = 1$ **to** M **do**
- 13 Sample a batch $\mathcal{X}_t^b$ from $\mathcal{X}_t$;
- 14 Forward the v^* and $\mathcal{X}_t^b$ through θ_v ;
- 15 Forward $\mathcal{X}_t^b$ through θ_t ;
- 16 Estimate the referenced uncertainty of $\mathcal{X}_t^b$ (Eq. (5));
- 17 Detect gap region $\mathcal{R}_t^b$ from $\mathcal{X}_t^b$ (Eq. (6));
- 18 Label gap region (Eq. (7));
- 19 Reducing gap region by optimizing $\mathcal{L}_{S-II}$ (Eq. (11)), updating target model θ_t .
- 20 Update target predictions in the prediction bank.
- 21 Set $v = v^*$.
- 22 **return** Adapted model θ_t .

the prompt context learned during the initial phase of task-specific customization. The function $I(\cdot, \cdot)$ corresponds to the mutual information function. The parameter α serves as a trade-off parameter, and the category balance term $L_B = \text{KL}(\bar{q} || \frac{1}{C})$ aligns with previous approaches (Tang et al., 2024; Yang et al., 2021), preventing solution collapse by ensuring the empirical label distribution $\bar{q}$ matches the uniform distribution $\frac{1}{C}$. For the reasons elaborated in Sect. 3.1, we also employ mutual information for alignment.

(c) *Gap region compression* Beyond semantics-guided reduction via $\mathcal{L}_{cac}$ and $\mathcal{L}_{pc}$, we further promote uncertainty-oriented gap-region compression by minimizing referenced entropy, formulated as:

$$\mathcal{L}_{rc} = \min_{\theta_t} \mathbb{E}_{x_i \in \mathcal{R}_t} U(x_i), \quad (10)$$

where m^k is the uncertainty margin used in the k -th epoch, its value is the same as the threshold in Eq. (6).

For better clarity, Fig. 3c shows how $\mathcal{L}_{rc}$ compresses the gap region. In Eq. (10), minimizing the loss enables the relatively easy subset within the current gap region (indicated by orange edge) to move beyond the current refer-

enced uncertainty contour m^k (shown in blue), entering the low uncertainty area (outside m^k). So, the relatively harder samples remain in the high-uncertainty zone (inside m^k), enabling the discovery of a more challenging gap region for the $(k + 1)$ -th epoch.

3.3 Objective and Model Training

To systematically distill and leverage task-specific knowledge from the ViL model, we adopt an epoch-wise training approach for DIFO++. The training process is divided into K epochs, each comprising two stages aligned with the two steps in the DIFO++ framework (Fig. 2). During the first stage, training is governed by the objective $\mathcal{L}_{S-I}$ (see Eq. (1)), and in the subsequent second stage, the objective function transitions to

$$\mathcal{L}_{S-II} = \mathcal{L}_{pc} + \beta \mathcal{L}_{cac} + \eta \mathcal{L}_{rc}, \quad (11)$$

where β and η are trade-off parameters. In Eq. (11), the component losses work with different functionality. $\mathcal{L}_{pc}$ also encourages interactive learning between the target and ViL models through mutual information. This design choice stems from the unsupervised nature of the Stage I ViL customization, which produces predictions with residual noise. Although this noise is partially mitigated, the outputs remain insufficiently reliable to serve as definitive ground truth. $\mathcal{L}_{rc}$ promotes a shift of the hard data from within to outside the contour as shown in Fig. 3c, whilst $\mathcal{L}_{pc}$ and $\mathcal{L}_{cac}$ jointly impose the guidance of moving direction to potential categories. For a clear view, we summarize the whole training procedure of DIFO++ in Algorithm 1.

4 Experimental Results

Datasets To evaluate DIFO++, we adopt four common SFDA benchmarks, as outlined below.

- (1) *Office-31* (Saenko et al., 2010) is a widely recognized small-scale dataset employed in visual domain adaptation. It encompasses three distinct domains: Amazon (A), Webcam (W), and Dslr (D), each capturing real-world objects situated within diverse office settings. In aggregate, the dataset comprises 4,652 images distributed across 31 categories.
- (2) *Office-Home* (Venkateswara et al., 2017) is a medium-scale dataset predominantly employed for domain adaptation. It amasses 15k images distributed across 65 categories, reflecting both working and family environments. The dataset introduces four unique domains: Artistic images (Ar), Clip Art (CL), Product images (Pr), and Real-world images (Rw).

Table 1 Closed-set SFDA accuracies (%) on the **Office-31** dataset

Method	ViL	A→D	A→W	D→A	D→W	W→A	W→D	Avg.
Source-model	–	79.1	76.6	59.9	95.5	61.4	98.8	78.6
SHOT (Liang et al., 2020)	✗	93.7	91.1	74.2	98.2	74.6	100	88.6
NRC (Yang et al., 2021)	✗	96.0	90.8	75.3	99.0	75.0	100	89.4
HCL (Huang et al., 2021)	✗	94.7	92.5	75.9	98.2	77.7	100	89.8
AaD (Yang et al., 2022)	✗	96.4	92.1	75.0	99.1	76.5	100	89.9
CoWA (Lee et al., 2022)	✗	94.4	95.2	76.2	98.5	77.6	99.8	90.3
SCLM (Tang et al., 2022)	✗	95.8	90.0	75.5	98.9	75.5	99.8	89.4
ELR (Yi et al., 2023)	✗	93.8	93.3	76.2	98.0	76.9	100	89.6
PLUE (Litrigo et al., 2023)	✗	89.2	88.4	72.8	97.1	69.6	97.9	85.8
TPDS (Tang et al., 2024)	✗	97.1	94.5	75.7	98.7	75.5	99.8	90.2
SF(DA) ² (Hwang et al., 2024)	✗	97.1	94.5	75.7	98.7	75.5	99.8	90.2
SiLAN (Wang et al., 2025)	✗	97.1	95.8	76.4	98.9	76.9	100	90.7
UCon-SFDA (Xu et al., 2025)	✗	94.8	95.4	98.9	100	77.1	77.1	90.6
SHOT-ViL (Liang et al., 2020)	✓	95.4	94.0	76.8	98.0	78.4	100	90.4
GKD-ViL (Tang et al., 2021)	✓	95.2	95.7	75.6	98.1	75.9	100	90.1
NRC-ViL (Yang et al., 2021)	✓	94.6	94.6	75.8	97.7	78.0	99.6	90.1
AaD-ViL (Yang et al., 2022)	✓	95.6	95.5	75.3	97.6	76.4	100	90.1
TPDS-ViL (Tang et al., 2024)	✓	95.0	93.8	77.3	98.6	77.2	99.8	90.3
Co-learn-V (Zhang et al., 2025)	✓	90.4	89.9	92.2	89.9	92.2	91.0	90.9
ViLAaD-V (Tarashima et al., 2025)	✓	97.6	93.8	79.6	96.7	79.7	100	91.2
DIFO (Tang et al., 2024)	✓	97.2	95.5	83.0	97.2	83.2	98.8	92.5
DIFO++ (Ours)	✓	97.2	96.9	82.9	97.2	83.3	99.0	92.8

SF means source-data-free; the best result is highlighted in bold

- (3) *VisDA* (Peng et al., 2017) presents a challenging landscape for domain adaptation with 12 synthetic-to-real transfer recognition tasks. The source domain houses 152k synthetic images, whereas the target domain encompasses 55k real object images sourced from Microsoft COCO.
- (4) *DomainNet-126* (Peng et al., 2019) is another large-scale dataset. As a subset of DomainNet containing 600k images of 345 classes from 6 domains of image styles, it has 145k images from 126 classes, sampled from 4 domains, Clipart (C), Painting (P), Real (R), Sketch (S), as (Saito et al., 2019) identify severe noisy labels in the dataset.

4.1 Implementation Details

Network structure The DIFO++ model contains two network branches. In the target model branch, the feature extractor consists of a deep architecture and a fully-connected layer followed by a batch-normalization layer. Same to the previous work (Liang et al., 2020; Long et al., 2018; Roy et al., 2022; Xu et al., 2019; Yang et al., 2021), the deep architecture is transferred from the deep models pre-trained on ImageNet (i.e., ResNet-50 is used on Office-31, Office-

Home and DomainNet-126, whilst ResNet-101 is adopted on VisDA). The ending classifier is a fully-connected layer with weight normalization. On the other hand, the ViL model branch chooses the most adopted CLIP as the implementation where the text encoder's transformer-based architecture follows modification proposed in (Radford et al., 2021) as the backbone. Regarding the image encoder, we adopt a single version corresponding to the implementation of DIFO (Tang et al., 2024). Specifically, in DIFO++, the image encoder adopts the ViT-B/32 architecture proposed in (CLIP Radford et al. 2021).

Parameter setting For the trade-off parameters α , β , and η , we set the same values on all datasets. Specifically, β and η in objective $\mathcal{L}_{S-II}$ (Eq. (11)) are 0.4 and 0.05, whilst α in $\mathcal{L}_{pc}$ (Eq. (9)) is 1.0. The parameter of Exponential distribution τ in Eq. (7) is specified to 10.0. The temperature parameters in Eq. (8) are $\tau = 0.1$. The number of the most-likely categories is set to $N = 2$. The parameters of uncertainty margin m^k is set to $(\epsilon, \gamma) = (0.01, 1.01)$.

Training setting We adopt the batch size of 64, SGD optimizer with momentum 0.9 and 15 training epochs on all datasets. The prompt template for initiation is the mostly used 'a photo of a [CLASS].' (Radford et al., 2021) where [CLASS] stands for the class name. All experiments are con-

Table 2 Closed-set SFDA result (%) on the **Office-Home** dataset

Method	ViL	Ar→Cl	Ar→Pr	Ar→Rw	Cl→Ar	Cl→Pr	Cl→Rw	Pr→Ar	Pr→Cl	Pr→Rw	Rw→Ar	Rw→Cl	Rw→Pr	Avg.
Source-model	–	43.7	67.0	73.9	49.9	60.1	62.5	51.7	40.9	72.6	64.2	46.3	78.1	59.2
SHOT (Liang et al., 2020)	✗	56.7	77.9	80.6	68.0	78.0	79.4	67.9	54.5	82.3	74.2	58.6	84.5	71.9
NRC (Yang et al., 2021)	✗	57.7	80.3	82.0	68.1	79.8	78.6	65.3	56.4	83.0	71.0	58.6	85.6	72.2
AaD (Yang et al., 2022)	✗	59.3	79.3	82.1	68.9	79.8	79.5	67.2	57.4	83.1	72.1	58.5	85.4	72.7
CoWA (Lee et al., 2022)	✗	56.9	78.4	81.0	69.1	80.0	79.9	67.7	57.2	82.4	72.8	60.5	84.5	72.5
SCLM (Tang et al., 2022)	✗	58.2	80.3	81.5	69.3	79.0	80.7	69.0	56.8	82.7	74.7	60.6	85.0	73.0
ELR (Yi et al., 2023)	✗	58.4	78.7	81.5	69.2	79.5	79.3	66.3	58.0	82.6	73.4	59.8	85.1	72.6
PLUE (Lirico et al., 2023)	✗	49.1	73.5	78.2	62.9	73.5	74.5	62.2	48.3	78.6	68.6	51.8	81.5	66.9
TPDS (Tang et al., 2024)	✗	59.3	80.3	82.1	70.6	79.4	80.9	69.8	56.8	82.1	74.5	61.2	85.3	73.5
SF(DA) ² (Hwang et al., 2024)	✗	57.3	79.1	81.6	68.4	78.2	78.7	69.5	56.5	81.9	74.0	59.9	85.1	72.5
SILAN (Wang et al., 2025)	✗	58.2	81.2	82.5	69.8	78.6	80.3	68.4	58.6	82.5	75.6	60.8	86.1	73.6
UCon-SFDA (Xu et al., 2025)	✗	65.6	87.8	91.0	78.6	79.3	87.6	80.2	65.9	87.3	83.2	69.1	88.7	80.3
SHOT-ViL (Liang et al., 2020)	✓	64.7	84.0	86.4	79.4	87.3	85.4	78.1	65.8	87.6	81.6	65.0	88.5	79.5
NRC-ViL (Yang et al., 2021)	✓	55.0	83.4	86.7	78.3	86.5	85.1	77.1	69.9	87.2	90.9	65.5	88.4	79.5
AaD-ViL (Yang et al., 2022)	✓	69.4	83.0	84.9	76.4	85.2	84.6	73.3	71.1	86.3	79.0	72.1	88.7	79.5
TPDS-ViL (Tang et al., 2024)	✓	67.7	82.7	86.1	76.5	88.0	86.3	73.6	69.4	87.1	79.3	70.5	89.7	79.7
Co-learn-V (Zhang et al., 2025)	✓	71.6	87.7	88.0	72.3	83.5	86.4	73.7	67.0	87.7	77.1	70.9	90.8	79.7
ViLAaD-V (Tarashima et al., 2025)	✓	66.6	86.8	87.5	77.2	88.1	87.1	77.1	65.9	88.5	78.7	68.0	89.1	80.0
DIFO (Tang et al., 2024)	✓	70.6	90.6	88.8	82.5	90.6	88.8	80.9	70.1	88.9	83.4	70.5	91.2	83.1
DIFO++ (Ours)	✓	73.2	91.4	89.7	83.6	91.0	89.8	80.8	73.8	90.0	84.3	74.2	92.3	84.5

ViL means whether using ViL model; the best result is highlighted in bold

Table 3 Closed-set SFDA result (%) on the **VisDA** dataset

Method	ViL	Plane	Bcycl	Bus	Car	Horse	Knife	Mcycl	Person	Plant	Sktrbrd	Train	Truck	Per-class
Source	–	60.7	21.7	50.8	68.5	71.8	5.4	86.4	20.2	67.1	43.3	83.3	10.6	49.2
SHOT (Liang et al., 2020)	✗	95.0	87.4	80.9	57.6	93.9	94.1	79.4	80.4	90.9	89.8	85.8	57.5	82.7
NRC (Yang et al., 2021)	✗	96.8	91.3	82.4	62.4	96.2	95.9	86.1	90.7	94.8	94.1	90.4	59.7	85.9
AaD (Yang et al., 2022)	✗	97.4	90.5	80.8	76.2	97.3	96.1	89.8	82.9	95.5	93.0	92.0	64.7	88.0
CoWA (Lee et al., 2022)	✗	96.2	89.7	83.9	73.8	96.4	97.4	89.3	86.8	94.6	92.1	88.7	53.8	86.9
SCLM (Tang et al., 2022)	✗	97.1	90.7	85.6	62.0	97.3	94.6	81.8	84.3	93.6	92.8	88.0	55.9	85.3
ELR (Yi et al., 2023)	✗	97.1	89.7	82.7	62.0	96.2	97.0	87.6	81.2	93.7	94.1	90.2	58.6	85.8
PLUE (Lirico et al., 2023)	✗	94.4	91.7	89.0	70.5	96.6	94.9	92.2	88.8	92.9	95.3	91.4	61.6	88.3
TPDS (Tang et al., 2024)	✗	97.6	91.5	89.7	83.4	97.5	96.3	92.2	82.4	96.0	94.1	90.9	40.4	87.6
SF(DA) ² (Hwang et al., 2024)	✗	96.8	89.3	82.9	81.4	96.8	95.7	90.4	81.3	95.5	93.7	88.5	64.7	88.1
SILAN (Wang et al., 2025)	✗	97.5	90.1	85.8	80.4	97.6	95.5	92.0	82.9	96.5	95.3	92.6	53.6	88.3
UCon-SFDA (Xu et al., 2025)	✗	98.4	90.7	88.6	80.7	97.9	96.9	93.1	83.8	97.6	95.9	92.6	59.1	89.6
SHOT-ViL (Liang et al., 2020)	✓	97.6	89.6	85.4	56.5	94.9	93.5	73.5	80.4	86.5	95.7	93.1	65.5	85.2
NRC-ViL (Yang et al., 2021)	✓	97.5	92.0	80.3	63.2	95.8	95.1	77.5	78.1	96.6	94.2	93.4	60.0	85.3
AaD-ViL (Yang et al., 2022)	✓	99.0	91.4	94.0	46.9	95.6	93.6	75.7	66.2	90.7	93.8	93.2	61.2	82.6
TPDS-ViL (Tang et al., 2024)	✓	98.5	91.6	89.2	85.5	97.4	12.3	88.7	50.1	88.2	92.0	92.8	48.9	78.0
Co-learn-V (Zhang et al., 2025)	✓	97.8	91.0	86.5	70.0	96.0	94.5	85.0	78.3	93.0	93.5	93.8	70.6	87.5
VILAaD-V (Tarashima et al., 2025)	✓	–	–	–	–	–	–	–	–	–	–	–	–	88.8
DIFO (Tang et al., 2024)	✓	97.5	89.0	90.8	83.5	97.8	97.3	93.2	83.5	95.2	96.8	93.7	65.9	90.3
DIFO++	✓	98.0	91.3	89.2	79.1	96.4	98.6	92.6	88.3	93.2	92.9	94.2	71.7	90.5

ViL means whether using ViL model; the best result is highlighted in bold

Table 4 Closed-set SFDA result (%) on the **DomainNet-126** dataset

Method	ViL	C→P	C→R	C→S	P→C	P→R	P→S	R→C	R→P	R→S	S→C	S→P	S→R	Avg.
Source	–	44.6	59.8	47.5	53.3	75.3	46.2	55.3	62.7	46.4	55.1	50.7	59.5	54.7
SHOT (Liang et al., 2020)	✗	63.5	78.2	59.5	67.9	81.3	61.7	67.7	67.6	57.8	70.2	64.0	78.0	68.1
NRC (Yang et al., 2021)	✗	62.6	77.1	58.3	62.9	81.3	60.7	64.7	69.4	58.7	69.4	65.8	78.7	67.5
AaD (Yang et al., 2022)	✗	60.6	75.2	58.0	58.2	79.7	54.6	63.3	67.8	54.6	70.6	66.8	76.4	65.5
CoWA (Lee et al., 2022)	✗	64.6	80.6	60.6	66.2	79.8	60.8	69.0	67.2	60.0	69.0	65.8	79.9	68.6
PLUE (Lirico et al., 2023)	✗	59.8	74.0	56.0	61.6	78.5	57.9	61.6	65.9	53.8	67.5	64.3	76.0	64.7
TPDS (Tang et al., 2024)	✗	62.9	77.1	59.8	65.6	79.0	61.5	66.4	67.0	58.2	68.6	64.3	75.3	67.1
SF(DA) ² (Hwang et al., 2024)	✗	61.5	75.3	59.6	67.8	83.5	61.6	68.8	70.5	60.2	68.5	67.7	75.1	68.3
SILAN (Wang et al., 2025)	✗	61.3	75.7	58.9	59.2	77.3	57.2	65.3	66.1	56.5	71.3	65.6	74.5	65.7
SHOT-ViL (Liang et al., 2020)	✓	72.8	84.5	72.0	75.1	85.2	72.2	78.2	75.5	71.6	75.2	73.6	83.2	76.6
NRC-ViL (Yang et al., 2021)	✓	73.7	84.4	73.6	76.1	85.4	73.6	80.4	76.9	73.9	74.9	74.8	82.1	77.5
AaD-ViL (Yang et al., 2022)	✓	72.2	84.2	72.0	75.3	85.3	72.6	78.2	77.4	73.6	74.8	74.7	82.8	76.9
TPDS-ViL (Tang et al., 2024)	✓	72.5	82.8	71.8	77.7	84.7	72.9	81.9	76.5	72.8	73.5	74.2	83.0	77.0
Co-learn-V (Zhang et al., 2025)	✓	78.2	89.2	76.2	82.7	89.2	76.3	82.5	78.0	76.0	82.5	78.3	89.2	81.5
DIFO (Tang et al., 2024)	✓	76.6	87.2	74.9	80.0	87.4	75.6	80.8	77.3	75.5	80.5	76.7	87.3	80.0
DIFO++	✓	78.1	89.7	76.5	81.7	89.6	77.2	82.3	80.2	75.9	82.3	79.4	89.7	81.9

ViL means whether using ViL model; the best result is highlighted in bold

Table 5 Results (%) of Partial-set SFDA (Top) and Open-set SFDA (Bottom) on the **Office-Home** dataset

Partial-set SFDA		Ar→Cl	Ar→Pr	Ar→Rw	Cl→Ar	Cl→Pr	Cl→Rw	Pr→Ar	Pr→Cl	Pr→Rw	Rw→Ar	Rw→Cl	Rw→Pr	Avg.
Source		45.2	70.4	81.0	56.2	60.8	66.2	60.9	40.1	76.2	70.8	48.5	77.3	62.8
SHOT (Liang et al., 2020)		64.8	85.2	92.7	76.3	77.6	88.8	79.7	64.3	89.5	80.6	66.4	85.8	79.3
HCL (Huang et al., 2021)		65.6	85.2	92.7	77.3	76.2	87.2	78.2	66.0	89.1	81.5	68.4	87.3	79.6
CoWA (Lee et al., 2022)		69.6	93.2	92.3	78.9	81.3	92.1	79.8	71.7	90.0	83.8	72.2	93.7	83.2
AaD (Yang et al., 2022)		67.0	83.5	93.1	80.5	76.0	87.6	78.1	65.6	90.2	83.5	64.3	87.3	79.7
CRS (Zhang et al., 2023)		68.6	85.1	90.9	80.1	79.4	86.3	79.2	66.1	90.5	82.2	69.5	89.3	80.6
DIFO (Tang et al., 2024)		70.2	91.7	91.5	87.8	92.6	92.9	87.3	70.7	92.9	88.5	69.6	91.5	85.6
DIFO++		75.9	90.2	90.9	88.4	91.8	92.5	87.5	75.6	93.3	89.5	74.3	90.8	86.7
Open-set SFDA		Ar→Cl	Ar→Pr	Ar→Rw	Cl→Ar	Cl→Pr	Cl→Rw	Pr→Ar	Pr→Cl	Pr→Rw	Rw→Ar	Rw→Cl	Rw→Pr	Avg.
Source		36.3	54.8	69.1	33.8	44.4	49.2	36.8	29.2	56.8	51.4	35.1	62.3	46.6
SHOT (Liang et al., 2020)		64.5	80.4	84.7	63.1	75.4	81.2	65.3	59.3	83.3	69.6	64.6	82.3	72.8
HCL (Huang et al., 2021)		64.0	78.6	82.4	64.5	73.1	80.1	64.8	59.8	75.3	78.1	69.3	81.5	72.6
CoWA (Lee et al., 2022)		63.3	79.2	85.4	67.6	83.6	82.0	66.9	56.9	81.1	68.5	57.9	85.9	73.2
AaD (Yang et al., 2022)		63.7	77.3	80.4	66.0	72.6	77.6	69.1	62.5	79.8	71.8	62.3	78.6	71.8
CRS (Zhang et al., 2023)		65.2	76.6	80.2	66.2	75.3	77.8	70.4	61.8	79.3	71.1	61.1	78.3	73.2
DIFO (Tang et al., 2024)		64.5	86.2	87.9	68.2	79.3	86.1	67.2	62.1	88.3	71.9	65.3	84.4	75.9
DIFO++		63.0	87.5	87.0	69.6	81.0	87.1	68.0	64.0	89.1	72.9	66.4	88.0	77.0

Table 6 Adaptation results (%) on **Office-Home** under the Continual SFDA setting

DIFO	Test					Test					Test					Test			
	Ar	Cl	Pr	Rw		Cl	Ar	Pr	Rw		Pr	Ar	Cl	Rw		Rw	Ar	Cl	Pr
Ar	92.6	41.4	67.6	77.3	Cl	93.8	60.1	60.6	69.3	Pr	94.8	58.4	40.7	78.0	Rw	94.3	68.7	48.7	79.3
Cl	77.2	64.1	63.3	72.2	Ar	78.0	78.2	64.2	76.6	Ar	86.7	73.7	43.5	80.0	Ar	89.7	79.0	45.3	75.9
Pr	72.4	64.7	84.9	76.8	Pr	74.3	68.3	85.4	77.5	Cl	75.7	70.0	61.8	76.4	Cl	79.4	74.0	64.3	70.9
Rw	74.9	60.6	83.1	83.3	Rw	70.1	66.7	84.5	83.7	Rw	83.8	68.7	55.1	83.0	Pr	82.1	69.6	60.1	87.2
↓	19.1	2.2	2.1	7.8	↓	21.0	13.7	1.1	11.9	↓	13.5	5.9	10.7	10.0	↓	11.2	9.1	6.5	8.9
DIFO++																			
Ar	92.6	<i>41.2</i>	<i>66.9</i>	<i>75.0</i>	Cl	93.8	60.5	61.3	70.4	Pr	94.8	<i>57.6</i>	<i>41.4</i>	<i>77.1</i>	Rw	94.3	<i>68.3</i>	<i>47.1</i>	<i>77.5</i>
Cl	79.5	66.1	62.8	72.5	Ar	79.6	76.5	64.9	76.8	Ar	88.5	77.0	50.5	81.2	Ar	89.7	<i>77.0</i>	45.5	<i>76.3</i>
Pr	75.0	65.1	86.9	78.9	Pr	77.9	68.3	87.6	78.7	Cl	79.3	70.0	66.4	<i>75.0</i>	Cl	82.3	<i>73.4</i>	65.7	71.4
Rw	77.8	63.2	84.9	83.7	Rw	71.8	67.3	84.7	84.4	Rw	85.6	73.4	59.7	86.9	Pr	82.6	70.1	60.4	86.0
↓	16.3	3.0	2.3	7.2	↓	18.5	11.4	3.3	11.1	↓	10.9	6.9	10.0	9.3	↓	10.0	6.8	8.1	8.3

The first column of each sub-table indicates the adaptation sequence. ↓ denotes the Relative Average Accuracy Drop (RD_{Acc}) of a test domain on the adaptation path compared with the performance when the domain is first seen. Formally, consider a sequence of accuracy results $a_i, a_{i+1}, \dots, a_{i+K}$ along an adaptation path. Here, a_i represents the initial accuracy when a domain is first encountered, and a_{i+k} denotes the test accuracy after the model has adapted to k subsequent domains; $RD_{Acc} = 1/K \sum_{k=1}^K (a_i - a_{i+k})/a_i \times 100\%$. In each sub-table, the final value in each row marked with ↓ represents the average of the three preceding values. bold font indicates the best mean of RD_{Acc} , while italic font denotes cases where the DIFO++ is outperformed by DIFO within the same adaptation flow

ducted with PyTorch on a single GPU of NVIDIA TITAN RTX.

4.2 Competitors

This paper validates the efficacy of our DIFO++ method through comparisons with 24 alternative methods, categorized into three distinct groups as outlined below.

- (1) The first group contains Source (the source model's results), CLIP (Radford et al., 2021), and Source+CLIP, where Source+CLIP directly averages the results of the source model and the CLIP model.
- (2) The second group includes 13 state-of-the-art SFDA models: SHOT (Liang et al., 2020), NRC (Yang et al., 2021), HCL (Huang et al., 2021), AaD (Yang et al., 2022), CoWA (Lee et al., 2022), SCLM (Tang et al., 2022), ELR (Yi et al., 2023), CRS (Zhang et al., 2023), PLUE (Litrico et al., 2023), TPDS (Tang et al., 2024), SF(DA)² (Hwang et al., 2024), SiLAN (Wang et al., 2025), and UCon-SFDA (Xu et al., 2025).
- (3) The third group includes 8 state-of-the-art SFDA models integrated the same CLIP model: DIFO (Tang et al., 2024), Co-learn-V (Zhang et al., 2025), ViLAaD-V (Tarashima et al., 2025), SHOT-ViL, GKD-ViL, NRC-ViL, AaD-ViL, and TPDS-ViL.

Among the third group, the comparisons with the suffix of “-ViL” are ViL versions of existing SFDA methods. In those updated versions, the CLIP's visual backbone (ViT

B/32) as the feature extractor, and theirs training process follows the original way. As such, all the compared methods benefit from the CLIP pre-trained knowledge in a design-generic manner, achieving a fair comparison.

4.3 Comparison on Closed-set SFDA setting

The results are listed in Tables 1, 2, 3, 4. DIFO++ surpasses the previous best method SiLAN (on Office-31 and Office-Home), PLUE (on VisDA), and GKD (on DomainNet-126) by **2.1%**, **10.9%**, **2.2%**, and **13.2%** in average accuracy, respectively. DIFO++ also improves DIFO by **0.3%** on Office-31, **1.4%** on Office-Home, **0.2%** on VisDA, and **1.9%** on DomainNet-126.

In particular, we observe over DIFO and DIFO++ that the datasets with more categories (Office-Home, DomainNet-126) benefit more than those with fewer categories (Office-31, VisDA). This result confirms the motivation of DIFO++: As the increasing number of categories tends to complicate the task specificity, the uniform attention on all data hinders DIFO from extracting useful information. This can be mitigated by the proposed gap region-driven transfer strategy.

In order to isolate the influence of model design, we compare with previous methods integrated with the same ViL (name ending with “-ViL”). DIFO++ still presents significant advantage, achieving at least gains of **2.4%**, **4.8%**, **2.3%** and **4.4%** on Office-31, Office-Home, VisDA, and DomainNet-126, respectively. This confirms the superiority of our design.

4.4 Comparison on Partial-Set and Open-Set Settings

Based on the results listed in Table 5, DIFO++ demonstrates significant performance improvements across adaptation settings. Compared with DIFO, DIFO++ outperforms by **1.1%** on both extending SFDA settings. Additionally, we observe that in the Partial-set SFDA setting (Top), DIFO++ achieves the best performance on 7 of 12 tasks, whereas in the more challenging Open-set SFDA setting (Bottom), it performs best on all transfer tasks as well, except for task Ar→Cl.

4.5 Continual SFDA and Source-Free Multi-target DA

In this part, we evaluate DIFO++ in the continual SFDA and source-free multi-target DA (SF-MTDA) setting, with experiment configurations in (Wang et al., 2022) and (Kumar et al., 2023), respectively. To evaluate the effect of DIFO++, DIFO is selected as a comparison, while the comparison method CoNMix (Kumar et al., 2023) serves for the SF-MTDA setting.

Based on the continual SFDA results in Table 6, we present two key observations. First, across all adaptation trajectories, DIFO++ consistently outperforms DIFO in 51 out of 64 cases (exceptions are marked in purple). This demonstrates that the relative advantage of DIFO++ remains robust throughout the continuous transfer process. Second, DIFO++ exhibits superior resistance to catastrophic forgetting compared to DIFO. This is evidenced by the performance in the most challenging scenarios (the second column of each sub-table) and its lower mean Relative Accuracy Drop (RD_{Acc}) for DIFO++, as shown in the final average of each ↓ row. Additionally, Table 7 shows that DIFO++ surpasses both DIFO and the conventional CoNMix by at least **2.5%**, further confirming its effectiveness in the SF-MTDA setting.

4.6 Comparison to the Original CLIP Model

The original CLIP model can conduct general image classification. We carry out a quantitative comparison between DIFO++ and CLIP, averaging the adaptation results of DIFO++ grouped by the target domain name.

As presented in the bottom of Table 8, DIFO++ outperforms CLIP on all tasks. On average accuracy, DIFO++ increases the performance by **13.0%**, **8.4%**, **7.6%** and **5.6%** in Office-31, Office-Home, VisDA and DomainNet-126, respectively. The result shows that the original CLIP model cannot fully excel to the target domain, and task-specific customization is needed.

Interestingly, compared with CLIP, except for VisDA with a tiny gap of **0.9%**, Source+CLIP averagely improve by **7.2%** at most on the other datasets. Meanwhile, Source+CLIP is beaten by DIFO++ with an increase of **6.2%**. These results

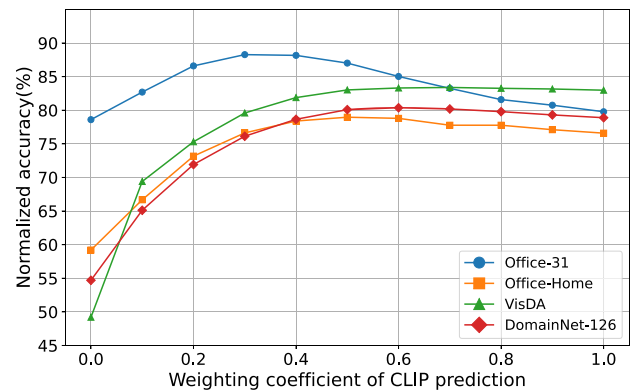


Fig. 5 The performance of the scheme directly weighting the source model and CLIP. All results are normalized by corresponding DIFO++ results for a clear view

imply that directly weighting the source model and CLIP is an intuitive knowledge adaptation scheme, but it is hard to perform adaptation deeply. Considering Source+CLIP is an average version, we conduct a comprehensive comparison with the weighting strategy where the weighting coefficient of CLIP prediction varies from 0.0 to 1.0. Here, we conduct this experiment based on more challenging CLIP due to its large performance gap with Source (see the first row in Tables 1, 2, 3, 4). For a clear view, all weighted accuracies are normalized by the corresponding DIFO++ accuracies, respectively. As shown in Fig. 5, no result can exceed the value of 1.0. This indicates that *weighting the source model and CLIP in a zero-shot manner cannot obtain desirable task-specific fusion, and a carefully designed distilling is necessary.*

4.7 Empirical Evidence on the Error-Prone of Conventional Methods

Due to the limited knowledge inherent in task-specific source models and unlabeled target data, the resulting pseudo-labels and auxiliary supervision are often contaminated with significant noise. Consequently, adaptation frameworks relying solely on these cues are inevitably error-prone. This challenge motivates us to incorporate a ViL model to bolster the adaptation process with more reliable, cross-modal knowledge.

Table 7 Accuracies (%) in SF-MTDA setting

Method	A→	C→	P→	R→	Avg.
CoNMix	75.6	81.4	71.4	73.4	75.4
DIFO	82.1	86.7	78.5	79.1	81.6
DIFO++	83.3	87.6	80.9	80.8	83.1

Table 8 Results (%) of CLIP and Source+CLIP on the four evaluation datasets

Method	Office-31				Office-Home					VisDA Sy→Re	DomainNet-126					Avg.
	→A	→D	→W	→Avg.	→Ar	→Cl	→Pr	→Rw	→Avg.		→C	→P	→R	→S	→Avg.	
CLIP	72.6	82.7	80.6	79.8	74.6	59.8	84.3	85.5	76.1	82.9	74.7	73.5	85.7	71.2	76.3	78.8
Source+CLIP	78.5	93.0	89.6	87.0	78.9	62.5	86.1	87.7	78.8	82.0	76.8	73.7	86.0	70.8	76.8	81.2
DIFO++	83.1	98.1	97.1	92.8	82.9	73.8	91.6	89.8	84.5	90.5	82.1	79.2	89.7	76.5	81.9	87.4

Table 9 Empirical evidence on the error-proneness of pseudo-label based supervision

Method	Ar→Cl		Cl→Ar		Avg.	
	PL-Acc	HC-Acc	PL-Acc	HC-Acc	PL-Acc	HC-Acc
Source-only	42.8	44.5	49.6	47.1	46.2	45.8
SHOT	55.2	57.3	68.9	72.2	62.1	64.7
TPDS	57.9	63.3	70.3	78.7	64.1	71.0
DIFO	64.6	80.6	77.7	81.3	71.2	80.9
DIFO++	66.0	84.7	80.1	89.7	73.1	87.2

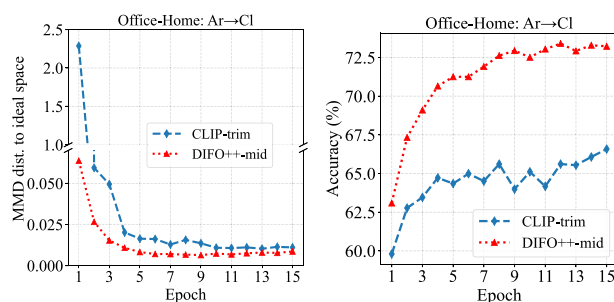
To validate our core motivation: ‘conventional methods are error-prone’, we conducted a comparative study on the symmetric transfer tasks $Cl \rightarrow Ar$ and $Ar \rightarrow Cl$ in Office-Home. The evaluation is based on two primary metrics: (1) pseudo-label accuracy (PL-Acc) and (2) the accuracy within the high-confidence subset (HC-Acc).

The experimental results are summarized in Table 9. Compared to DIFO and DIFO++, baseline methods such as SHOT and TPDS, which do not leverage ViL knowledge, exhibit substantially higher levels of predictive noise across the target domain, as evidenced by their lower PL-Acc. More compellingly, even within their high-confidence subsets, these traditional methods yield a notably lower HC-Acc than our proposed frameworks. These findings suggest that conventional approaches are inherently more prone to miscalibration and error. Furthermore, the performance gap indicates that integrating an external ViL model can effectively transcend the epistemic boundaries of task-specific source models and unlabeled target data, providing more robust guidance for adaptation.

4.8 Task-Specific Knowledge Adaptation Analysis

In this part, we give a feature space shift analysis using the measure of MMD (Maximum Mean Discrepancy) distance (Zongxianlee, 2019) to verify whether the proposed method ensures a task-specific knowledge adaptation.

In this experiment, we first train a domain-invariant Oracle model over all Office-Home data with real labels, and use the logits to express the ideal task-specific space $\mathcal{O}$. After that, an analysis is conducted on the transfer task $Ar \rightarrow Cl$. During this adaptation, there are T (epoch number) intermediate target models and customized CLIP models. We feedforward the target data through each intermediate model and take the log-

**Fig. 6** The evolving dynamics of MMD distance during adaptation of $Ar \rightarrow Cl$ in Office-Home. **Left** and **Right** present the varying curves of MMD distance and accuracy, respectively

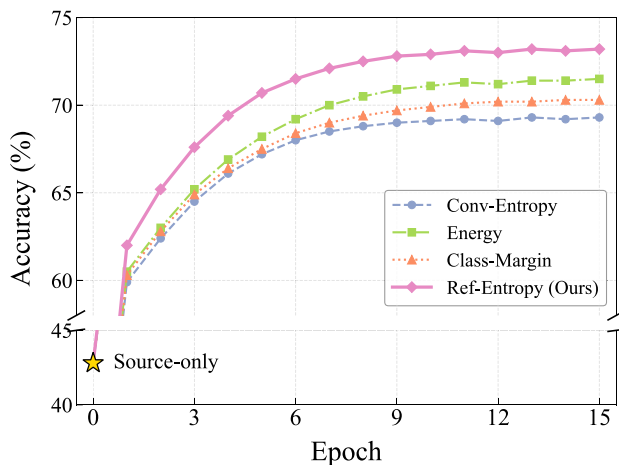
its as a space. Thus, we obtain T intermediate target feature spaces $\{\mathcal{U}_k\}_{k=1}^T$ and T intermediate customized CLIP feature spaces $\{\mathcal{V}\}_{k=1}^T$. Within this context, these intermediate spaces can depict the task-specific distillation to $\mathcal{O}$.

In the left of Fig. 6, we give the MMD distance change curve of $\{\mathcal{U}_k\}_{k=1}^T$ (in red, termed DIFO++-mid) and $\{\mathcal{V}\}_{k=1}^T$ (in blue, termed CLIP-trim), taking $\mathcal{O}$ as the original space. It is seen that at early epochs (1~4), DIFO++-mid and CLIP-trim sharply decrease and then maintain a gradual decrease in the following epochs. Meanwhile, this change is consistent with the accuracy varying shown in the right of Fig. 6.

These results indicate that our DIFO++ encourages task-specific knowledge adaptation due to the convergence on the ideal task-specific space. Besides, we observe two details. First, after epoch 1, CLIP-trim’s distance reduces by **2.2**, which is **59.5** times of DIFO++-mid’s decrease of **0.037**. This is because CLIP represents a heterogeneous space of vision-language, much different from the vision space $\mathcal{O}$. Furthermore, the large decrease in distance confirms the effect of customization. Second, the synchronized distance

Table 10 Ablation study in Closed-set setting (%)

#	$\mathcal{L}_{S-I}$	$\mathcal{L}_{S-II}$			Office-31	Office-Home	VisDA	Avg.
		$\mathcal{L}_{cac}$	$\mathcal{L}_{pc}$	$\mathcal{L}_{rc}$				
1	X	X	X	X	78.6	59.2	49.2	62.3
2	✓	X	X	X	82.4	77.4	84.4	81.4
3	X	✓	X	X	82.1	76.5	88.6	82.4
4	X	X	✓	X	90.7	77.6	88.6	85.6
5	X	X	X	✓	76.7	60.1	54.1	63.6
6	✓	✓	✓	X	92.5	83.1	90.3	88.6
7	✓	✓	X	✓	86.5	68.7	90.0	81.7
8	✓	X	✓	✓	83.4	70.1	88.6	80.7
9	X	✓	✓	✓	91.3	82.1	89.8	87.7
10	✓	✓	✓	✓	92.8	84.5	90.5	89.3
11	DIFO++ w/ KL				87.5	75.4	85.4	82.8
12	DIFO++ w/ RS				90.9	83.3	89.4	87.8
13	DIFO++ w/ p'_i				89.0	73.7	87.1	83.3
14	DIFO++ w/ p''_i				90.6	83.5	88.7	87.6
15	DIFO++ w/ ENT				91.9	83.0	88.9	87.9

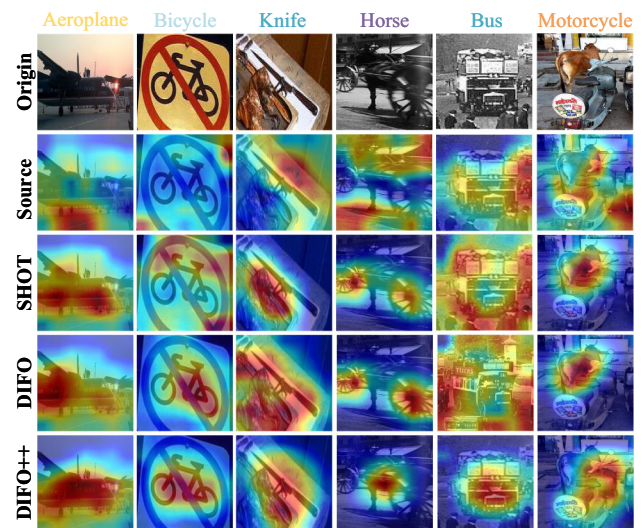
**Fig. 7** Convergence curves of different uncertainty metrics on the Ar→CI task in Office-Home

reductions of CLIP-trim and DIFO++-mid indicate that the interaction between the target model and CLIP is a crucial design for task-specific distillation.

4.9 Ablation Study

Effect of designs in DIFO++. In this part, we evaluate in five aspects. (1) Effect of the proposed objectives: $\mathcal{L}_{S-I}$, ones in $\mathcal{L}_{S-II}$ including $\mathcal{L}_{cac}$, $\mathcal{L}_{pc}$, and $\mathcal{L}_{rc}$, (2) effect of optimization of mutual information, (3) effect of uncertainty-aware sampling, (4) effect of task-specific customization, and (5) effect of weighting-based gap-region labeling.

In the first issue, we isolate the effect of each component by introducing or removing it singly. As shown in the first

**Fig. 8** Visualization of six classes from VisDA

5 rows of Table 10, there are performance improvements to some extent compared with the baseline of the source model (row 1) when the four losses work alone. In particular, the 1.3% gain of $\mathcal{L}_{rc}$ (row 5) confirms the rationale of identifying confident data. Additionally, removing any loss component leads to performance deterioration (rows 6–11). When $\mathcal{L}_{cac}$ or $\mathcal{L}_{pc}$ are unavailable, the average accuracy drops the most (by about 9%). The phenomenon is consistent with the cases in rows 3–4: as $\mathcal{L}_{cac}$ or $\mathcal{L}_{pc}$ work alone, the accuracy significantly improves. This observation confirms the effect of our designs: encouraging category attention calibration, paired with predictive consistency.

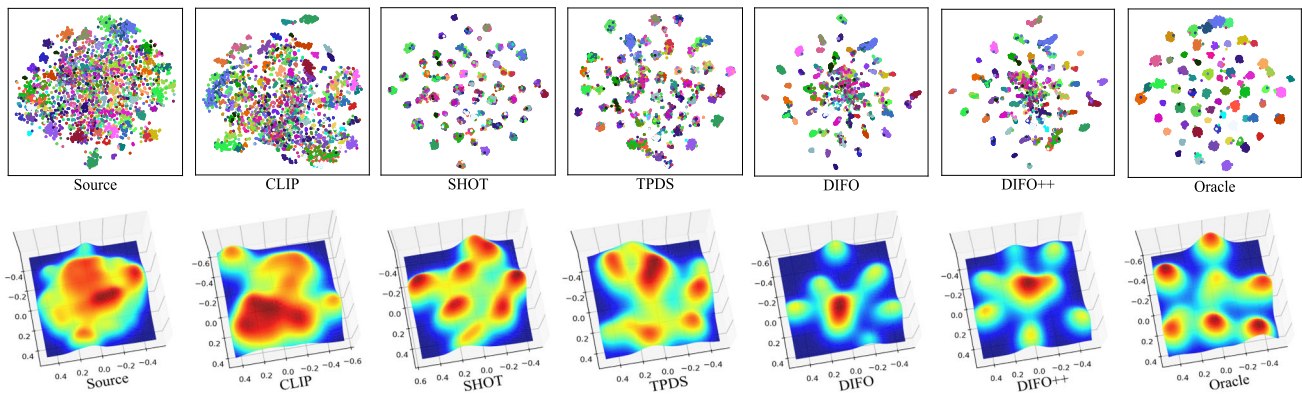


Fig. 9 Feature distribution visualization on transfer task Ar→Cl in Office-Home. Oracle: the model trained on target domain Cl using ground-truth labels. Different colors stand for different categories. **Top:**

t-SNE feature distribution over 65 categories. **Bottom:** The corresponding 3D density charts. For clarity, the first 10 categories were used in this plot

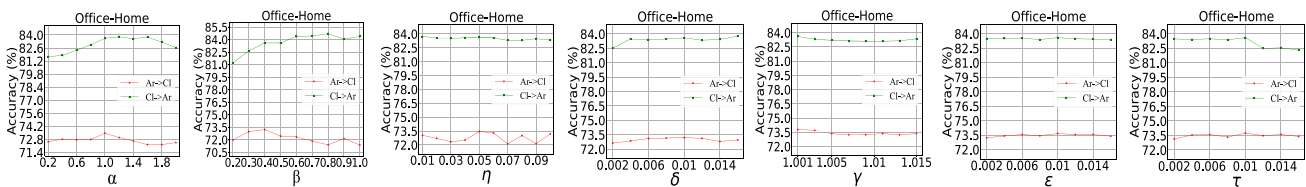


Fig. 10 Sensitivity of the hyper-parameters α , β , η , δ , γ , ϵ , and τ

For the other four issues, we propose three variations of DIFO++ to accomplish effect evaluation. The first one is DIFO++ w/ KL, where the mutual information maximization loss in $\mathcal{L}_{S-I}$ and $\mathcal{L}_{pc}$ are replaced by KL divergence loss. The second one is DIFO++ w/ RS, where referenced entropy-based hard data sampling is changed to a random selection. The third one is the same as the loss ablation case in row 9, where the prompt learning-based customization for CLIP is canceled, and the input prompt is set to a fixed template (“a photo of a [CLS]”). The last one includes DIFO++ w/ p'_i and DIFO++ w/ p''_i , which substitute the weighted combination (Eq. (7)) with either the individual target prediction p'_i or the ViL prediction p''_i . As shown in rows 11~14 of Table 10, DIFO++ outperforms all three comparisons, with an average improvement of at least **1.5%**, confirming the effect of adopting mutual information optimization, task-specific customization, and referenced uncertainty, respectively.

Effect of referenced entropy We introduce a variant, DIFO++ w/ ENT, which substitutes our referenced uncertainty with conventional information entropy. As reported in Row 15 of Table 10, this substitution results in a performance drop of **1.4%**. Furthermore, we conduct a comparative analysis against established uncertainty metrics, including conventional entropy, Energy-based scores (Liu et al., 2020), and Class-Margin (Song et al., 2024), under an identical experimental configuration as DIFO++. As illustrated in Fig. 7, our referenced entropy (Ref-Entropy) exhibits a superior

convergence profile throughout the adaptation phase. These empirical results underscore the effectiveness of Ref-Entropy in providing more stable and robust guidance for our curriculum-learning-based knowledge distillation.

4.10 Further Model Analysis

Grad-CAM visualization To better understand our DIFO++ method, we present a Grad-CAM visualization comparison between the source model, the state-of-the-art SFDA methods SHOT, DIFO, and our DIFO++. Specifically, as shown in Fig. 8, for images containing a single object (see columns 1 to 3), DIFO++ accurately focuses on the discriminative visual part, whereas other methods distribute attention across the entire image. For images containing multiple objects (see columns 4 to 6), DIFO++ shows superior alignment with the target semantics described by the ground truth labels, compared to other methods, which often misdirect attention to incorrect objects. These findings highlight the effectiveness of DIFO++ in combining the domain generality of ViL models with the task-specific adaptability of source models.

Feature distribution visualization To visualize feature distribution, we conducted a toy experiment on the Ar→Cl task in the Office-Home dataset, using the t-SNE tool. We include five comparison models: CLIP’s zero-shot (labeled as CLIP), SHOT, TPDS, DIFO, and Oracle (trained on domain Cl with real labels). As shown at the top of Fig. 9, from CLIP to

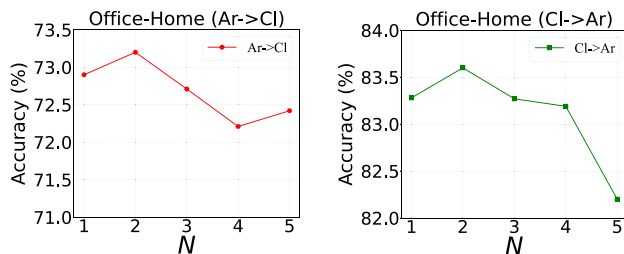


Fig. 11 Sensitivity of parameter N

DIFO++, category aliasing gradually diminishes. Compared to Oracle, DIFO++ exhibits the most similar distribution shape. In particular, the mixture degree of multi-class in the middle zone is weaker than that of DIFO. The observation offers an explanation for the advantage of DIFO++ over DIFO. To further validate this observation, we present the 3D density chart results at the bottom of Fig. 9. These results confirm the effectiveness of DIFO++ in terms of feature distribution.

Sensitivity to hyper-parameters In DIFO++, α in objective loss L_{PC} (Eq. (9)) and (β, η) in $\mathcal{L}_{S-II}$ (Eq. (11)), are the trade-off parameters. δ in Eq. (4) and τ in Eq. (7) are weighted parameters, while threshold parameters (ϵ, γ) in Eq. (6) govern the sample selection process within the curriculum learning framework. We test their sensitivity of performance on the symmetric transfer tasks $Cl \rightarrow Ar$ and $Ar \rightarrow Cl$ in Office-Home. Figure 10 shows no significant drops of accuracy from varying these parameters, suggesting good stability.

In addition, we investigate the sensitivity of the hyperparameter N in $\mathcal{L}_{cac}$, as illustrated in Fig. 11. We observe that performance peaks at $N = 2$ and subsequently declines as N increases. This indicates that although considering multiple top categories is beneficial, a high value of N leads to the inclusion of excessive noise. Based on our empirical findings, $N = 2$ serves as a robust choice across different scenarios.

Dynamics of gap-region reduction To validate that our method effectively encourages gap-region reduction, we epoch-wise visualize the adaptation using the t-SNE tool. As illustrated in Fig. 12, at Epoch 0 (representing the source model results), the feature space exhibits several isolated clusters surrounded by a large number of samples forming a mist-like distribution, which constitutes the 'gap-region'.

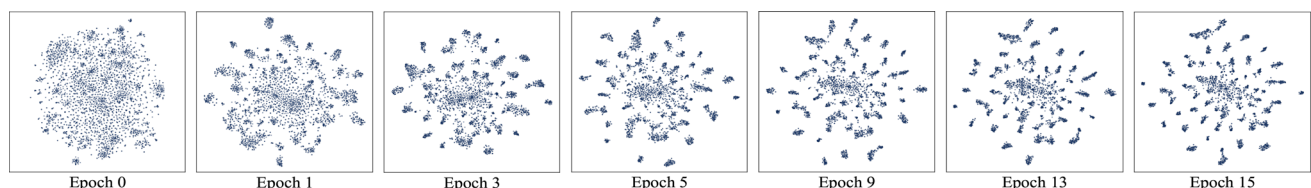


Fig. 12 Dynamics of gap-region reduction in epoch-wise view

Table 11 Resource demands on $Ar \rightarrow Cl$ task

#	Item	SHOT	TPDS	DIFO	DIFO++
1	GPU Mem. cost $\downarrow$ (G)	7.868	8.535	10.164	10.060
2	Time per iter. $\downarrow$ (s)	0.407	0.742	0.929	0.903

Throughout the adaptation process, these diffuse features are progressively eliminated, resulting in clear semantic clustering. Such a transition confirms the efficacy of our approach in driving gap-region reduction and feature alignment.

Resource demands In this part, we evaluate the resource demands of DIFO++, as reported in Table 11. Regarding GPU memory consumption, DIFO and DIFO++ require an additional 2 GB compared to the baseline SHOT and TPDS. This overhead is primarily attributed to the integration of the external Vision-Language (ViL) model. However, the computational latency per iteration does not increase substantially; compared to SHOT and TPDS, our methods only introduce an additional 0.5 s and 0.2 s, respectively, due to ViL inference and prompt learning. Notably, DIFO++ exhibits lower overall resource demands than DIFO. This efficiency gain is expected, as the curriculum learning strategy adopted in DIFO++ gradually reduces the effective scale of training data throughout the adaptation process.

Convergence analysis In Fig. 13, we illustrate the convergence behavior of DIFO++ on the VisDA dataset throughout the adaptation phase. For benchmarking, we compare our method against SHOT, AaD, and TPDS. It is evident that the area under the convergence curve for DIFO++ is significantly larger than that of the competitors. This result demonstrates that DIFO++ achieves superior classification performance with an accelerated convergence rate, reaching a higher steady-state accuracy in fewer iterations.

5 Conclusion

In this work, we introduce a novel DIFO++ approach to address the SFDA problem. To the best of our knowledge, this marks the initial endeavor to address SFDA by leveraging a pre-trained ViL foundation model, departing from previous approaches that predominantly concentrated on self-mining auxiliary information. DIFO++ is featured with alternating

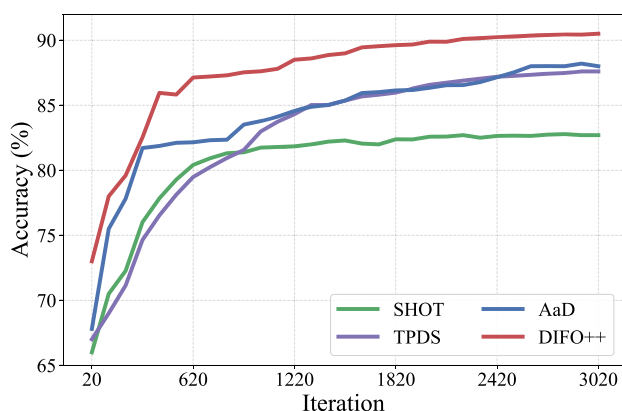


Fig. 13 Convergence behavior comparison on VisDA during the adaptation phase

between customization of the ViL model and the transfer of task-specific knowledge from the customized ViL model. Our approach involves three pivotal designs: (1) a mutual information-based alignment for ViL customization, (2) a gap region-driven knowledge adaptation for transferring the customized ViL knowledge, and (3) a referenced entropy for uncertainty evaluation. Extensive experiments demonstrate that DIFO++ outperforms the state-of-the-art alternatives across multiple challenging benchmarks.

Limitations While DIFO++ demonstrates the potential of leveraging multimodal foundation models for the SFDA problem, it has several limitations. First, its ViL model customization is based on prompt learning, which requires access to the model itself. As a result, DIFO++ cannot be deployed in black-box or cloud-based settings where model access is restricted. Second, DIFO++ lacks an anti-forgetting mechanism, making it unsuitable for continuous learning scenarios. Third, the referenced entropy and gap region labeling depend on historical information accumulated during adaptation, limiting its applicability in online settings such as Test-Time Adaptation (TTA). Extending DIFO++ to address these challenging scenarios presents an important direction for future research.

A. A Proof of Lemma 1

Restatement of Lemma 1 Given two random variables X , Y . Their mutual information $I(X, Y)$ and KL divergence $D_{KL}(X||Y)$ satisfy the unequal relationship as follows.

$$-I(X, Y) \leq D_{KL}(X, Y). \quad (12)$$

Proof Suppose the probability density function (PDF) of X and Y are $p(x)$ and $p(y)$, respectively; their joint PDF is

$p(x, y)$. We have

$$\begin{aligned} I(X, Y) &= \sum p(x, y) \log \frac{p(x, y)}{p(x) \cdot p(y)} \\ &= D_{KL}(p(x, y) || p(x) \cdot p(y)). \end{aligned}$$

Well known, the KL divergence is non-negative (Eguchi & Copas, 2006). Thus,

$$-I(X, Y) \leq 0 \leq D_{KL}(X||Y).$$

□

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Data Availability The authors confirm that the data supporting the findings of this study are available within the articles: (Saenko et al., 2010) (Office-31), (Venkateswara et al., 2017) (Office-Home), (Peng et al., 2017) (VisDA), and (Peng et al., 2019) (DomainNet-126).

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical Approval Ethical approval was not required for this study as it involves the analysis of publicly available datasets and does not involve human participants or animals.

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